

EASWARI ENGINEERING COLLEGE

(An Autonomous Institution)

(A Unit of SRM Group of Educational Institutions)

DEPARTMENT OF BIOMEDICAL ENGINEERING

2026-2027 ODD SEMESTER

To

The Organizing Committee

36-Hour AI Hackathon

Tata Centre for Artificial Intelligence and Machine Learning

National Institute of Technology Tiruchirappalli

Subject: Letter of Endorsement for the Project "PolypNet-3D: Temporally-Consistent Deep Learning System for Real-Time Polyp Detection in Colonoscopy Video"

Dear Organizing Committee,

It is with great pleasure that I endorse the participation of Ms. Jayasri K and her team from the Department of Biomedical Engineering and Department of Artificial Intelligence, Easwari Engineering College, in the 36-Hour AI Hackathon organized by the Tata Centre for Artificial Intelligence and Machine Learning, National Institute of Technology Tiruchirappalli.

Our team consists of the following members:

1. **K. Jayasri** (Register No.: 310624121022) - Department of Biomedical Engineering
2. **Varsha S Rao** (Register No.: 310624121058) - Department of Biomedical Engineering
3. **Gokul R** (Register No.: 310624148027) - Department of Artificial Intelligence and Machine Learning

The team has proposed an innovative healthcare solution titled "PolypNet-3D," which resolves the issue of frame-to-frame flickering in AI-assisted polyp detection during live colonoscopies. By combining real-time object detection with temporal stabilization, the system eliminates erratic detection loss across consecutive video frames, delivering a continuous, stable, and clinically trustworthy visual overlay to assist gastroenterologists during procedures.

Viability of the Idea

Colorectal cancer screening relies heavily on timely and accurate polyp detection during colonoscopy, yet most existing AI-assisted detection tools treat the procedure as a sequence of unrelated static images, leading to inconsistent, flickering detections that clinicians find difficult to trust in real time. PolypNet-3D directly addresses this gap by treating colonoscopy as a continuous video procedure and applying temporal consistency techniques to ensure stable, reliable detection throughout. Built entirely on well-established medical imaging datasets, the solution is technically sound, reproducible, and scalable. By improving the real-world usability of AI-assisted polyp detection, this project has the potential to support earlier and more consistent identification of precancerous lesions, contributing meaningfully to colorectal cancer screening infrastructure.

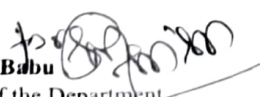
Ability to Implement

Ms. K. Jayasri and her team have consistently demonstrated strong technical aptitude and hands-on experience in applying deep learning and computer vision techniques to biomedical imaging challenges, including prior work in AI-based diagnostic pipelines for cardiac and oncological screening. Their combined academic foundation across Biomedical Engineering and Artificial Intelligence, along with practical experience in model development, dataset handling, and healthcare-focused problem framing, equips them with the technical capability required to design, train, and deploy the proposed detection and temporal-smoothing pipeline within the 36-hour hackathon timeframe. I am confident that the team possesses the collaborative mindset, technical rigor, and determination necessary to execute this project effectively and deliver a working, demonstrable solution.

I wholeheartedly endorse their participation in this prestigious hackathon and believe that their project has the potential to make a meaningful contribution towards advancing accessible, technology-driven healthcare solutions for society.

I wish the team every success in the competition and appreciate your consideration of their application.

Yours sincerely,


Dr. G. Babu

Head of the Department

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